

CHAPTER C1 GENERAL

C1.1 SCOPE

The minimum design loads, hazard levels, associated criteria, and intended performance goals contained in this standard are derived from research and observed performance of buildings, other structures, and their nonstructural components under the effects of loads. These parameters vary depending on the relative importance of the building, other structure, or nonstructural component. The loads provided in this standard include loads from both normal operations and rare hazard events. All loads and associated criteria are prescribed to achieve an intended performance, which is defined by a reliability index or limit state exceedance probability or preservation of function during a specified hazard event.

Loads and load combinations are set forth in this document with the intent that they be used together. If one were to use loads from some other source with the load combinations set forth herein or vice versa, the reliability of the resulting design may be affected.

With the 2016 edition of the standard, the title was modified to include the words “and Associated Criteria” to acknowledge what has been in this standard for many editions. For example, earthquake loads contained herein are developed for structures that possess certain qualities of ductility and postelastic energy dissipation capability. For this reason, provisions for design, detailing, and construction are provided in Chapters 11 through 22. In some cases, these provisions modify or add to provisions contained in design specifications. However, this standard only adds associated criteria when the modification is needed to achieve the intended structural performance when subjected to the loads specified herein.

C1.3 BASIC REQUIREMENTS

C1.3.1 Strength and Stiffness. Buildings and other structures must satisfy strength limit states in which members and components are proportioned to safely carry the design loads specified in this standard to resist buckling, yielding, fracture, and other unacceptable performance. This requirement applies not only to structural components but also to nonstructural elements, the failure of which could pose a substantial safety or other risk. Chapter 30 of this standard specifies wind loads that must be considered in the design of cladding. Chapter 13 of this standard specifies earthquake loads and deformations that must be considered in the design of nonstructural components and systems designated in that chapter.

Although strength is a primary concern of this section, strength cannot be considered independent of stiffness. In addition to considerations of serviceability, for which stiffness is a primary consideration, structures must have adequate stiffness to ensure stability. In addition, the magnitude of load imposed on a

structure for some loading conditions, including earthquake, wind, and ponding, is a direct function of the structure’s stiffness.

Another important consideration related to stiffness is damage to nonstructural components resulting from structural deformations. Acceptable performance of nonstructural components requires either that the structural stiffness be sufficient to prevent excessive deformations or that the components can accommodate the anticipated deformations.

Standards produced under consensus procedures and intended for use in connection with building code requirements contain recommendations for resistance factors for use with the strength design procedures of Section 1.3.1.1 or allowable stresses (or safety factors) for the allowable stress design procedures of Section 1.3.1.2. The resistances contained in any such standards have been prepared using procedures compatible with those used to form the load combinations contained in Sections 2.3 and 2.4. When used together, these load combinations and the companion resistances are intended to provide reliabilities approximately similar to those indicated in Tables 1.3-1, 1.3-2, and 1.3-3. Some standards known to have been prepared in this manner include the following:

ACI 318, *Building Code Requirements for Structural Concrete*, American Concrete Institute.

AISC 341, *Seismic Provisions for Structural Steel Buildings*, American Institute of Steel Construction.

AISC 358, *Prequalified Connections for Special and Intermediate Steel Moment Frames for Seismic Applications*, American Institute of Steel Construction.

AISC 360, *Specification for Structural Steel Buildings*, American Institute of Steel Construction.

AISI S100-16, *North American Specification for the Design of Cold-Formed Steel Structural Members*, American Iron and Steel Institute.

Aluminium Association, *Specification for Aluminum Structures*, Aluminum Association.

AWC NDS-2015, *National Design Specification for Wood Construction*, American Wood Council.

AWC SDPWS-2015, *Special Design Provisions for Wind and Seismic*, American Wood Council.

SEI/ASCE 8, *Specification for the Design of Cold-Formed Stainless Steel Structural Members*, ASCE.

TMS 402, *Building Code Requirements and Specification for Masonry Structures*, The Masonry Society.

TMS 602, *Specification for Masonry Structures*, The Masonry Society.

C1.3.1.3 Performance-Based Procedures. Section 1.3.1.3 introduces alternative performance-based procedures that may be used in lieu of the procedures of Sections 1.3.1.1 and 1.3.1.2

to demonstrate that a building or other structure, or parts thereof, has sufficient strength. These procedures are intended to parallel the so-called “alternative means and methods” procedures that have been contained in building codes for many years. Such procedures permit the use of materials, design, and construction methods that differ from the prescriptive requirements of the building code, or in this case the standard, that can be demonstrated to provide equivalent performance. Such procedures are useful in that they permit innovation and the development of new approaches before the building codes and standards have an opportunity to provide for these new approaches. In addition, these procedures permit the use of alternative methods for certain special structures, which may not be covered by code but by means of their occupancy, use, or other features, can provide acceptable performance without compliance with the prescriptive requirements.

The reliability of a proposed design does not need to be evaluated when the standard’s design procedures in Sections 1.3.1.1 and 1.3.1.2 are applied. However, when performance-based procedures are used, the reliability achieved for the proposed design should be consistent with the target reliabilities stipulated in Section 1.3.1.3.

Alternative design methods have a range of implementation levels. Such methods are addressed in standards or best practice guidelines that address performance-based design goals and methodologies that incorporate the fundamental basis of reliability analysis. A minimum level of alternative design would involve standard design procedures, with analyses based on the requirements of Section 1.3.1.3. For example, a building that exceeds the code limits for building height could be designed with applicable codes and standards, and member demand and capacities would be checked to determine their adequacy for the design loads and conditions. For seismic design, the provisions of the ASCE 41 standard and of the *Tall Buildings Initiative, Guidelines for Performance-Based Seismic Design of Tall Buildings* (PEER 2010) were either calibrated by structural performance level or were demonstrated in comparison with prescriptive design methods to provide reliabilities equal to or better than Table 1.3-2.

Flood load factors in ASCE 7 may not achieve the reliability targets of Table 1.3-1. For structures within the 100-year coastal flood zone, the load factor of 2.0 was based on a beta value of 2.5 (Mehta et al. 1998) rather than 3.0. For structures not in the coastal zone, the load factor of 1.0 reflects the prescriptive minimum 100-year flood elevation for stillwater flooding; thus this flood has a 1% annual chance of being exceeded, which is essentially a beta of 1.3. For Risk Category III and IV structures, no reliability analysis has been performed. For storm events of greater return period, the flood hazard expands both in spatial extent and in depth. For those structures and components of structures that were not subject to the prescriptive flood design requirements, any design resistance depends on that imparted from design for other hazards.

Tsunami and extraordinary events do not have specific reliability tables in this section. Guidance for performance-based design of tsunami can be found in Chapter 6 and its associated commentary. For extraordinary events, the user is directed to Section C2.5 for discussion on reliability consideration for extraordinary events.

The alternative procedures of Section 1.3.1.3 may be used to demonstrate adequacy for one or more design loads, while the standard procedures of Sections 1.3.1.1 and 1.3.1.2 are used to demonstrate adequacy for other design loads. For example, it is relatively common to use the alternative procedures to demonstrate adequate earthquake, fire, or blast resistance, while the

standard prescriptive procedures of Sections 1.3.1.1 and 1.3.1.2 are used for all other loading considerations.

The alternative procedures of Section 1.3.1.3 are intended to be used in the design of individual projects, rather than as the basis for broad qualification of new structural systems, products, or components. Procedures for such qualification are beyond the scope of this standard, as the limited number of test data required in Section 1.3.1.3.2 are not appropriate for the application of new materials in a structural system or the development of prefabricated structural assemblies intended for general widespread use in structural systems. A more robust level of testing is needed for new materials or structural assemblies for general use.

Section 1.3.1.3 requires demonstration that a design has adequate strength to provide an equivalent or lower probability of failure under load than that adopted as the basis for the prescriptive requirements of this standard for buildings and structures of comparable Risk Category. Tables 1.3-1, 1.3-2, and 1.3-3 summarize performance goals, expressed in terms of target reliabilities, associated with protection against structural failure that approximate those notionally intended to be achieved using the design procedures of Section 2.3. The target reliability indices are provided for a 50-year reference period, and the probabilities of failure are annual probabilities. Annualized probabilities can be applied to limit states where the loads and member resistance do not vary over the reference period. If a member is subject to degradation, such as corrosion, then degradation effects over the service period should be considered through a time-varying or stochastic process as part of the reliability analysis.

The target reliabilities have been developed and vetted by a number of consensus groups over a period of more than 30 years and have been confirmed through professional practice in AISC 360, ACI 318, and other standards and documents. The target reliabilities for Risk Category II in Table 1.3-1 are based on probabilistic analyses of structural member performance for strength design procedures and are documented in Ellingwood et al. (1980, 1982) and Galambos et al. (1982). The reliabilities are consistent with those adopted by the NBCC (2010), CEN 250 (2002) and ISO (1998). Structural members and connections designed using typical specifications for engineering materials (steel, reinforced concrete, masonry, timber) were analyzed to determine their reliabilities for common limit states, such as yielding in tension members, formation of plastic hinges in compact laterally supported beams, or column buckling and connection fracture for a nominal service period of 50 years. The reliabilities were determined for load combinations involving dead, live, wind, snow, and earthquake loads initially. The target reliabilities listed in Table 1.3-1 are based on strength criteria for structural members. The target reliabilities listed in Table 1.3-2 for strength and deflection limit states are based on strength and deflection criteria for system response to earthquakes where inelastic behavior is assumed. The target reliabilities for Risk Categories I, III, and IV were determined by reviewing the intended performance of structural members and systems, as well as target reliabilities specified by other codes and standards for similar performance criteria.

Seismic design practice has evolved in the past three decades from the original reliability basis mentioned previously. The target reliabilities in Table 1.3-2 for earthquake-resistant structural system design are defined for the response of the structural system as described in NIST (2012), which was prepared by NEHRP Consultants Joint Venture: For Risk Category I and Risk Category II structures, i.e., $I_e = 1.0$, acceptable Life Safety risk is defined by an “absolute” collapse probability of 1% in 50 years and a “conditional” probability of 10% given MCE_R ground

motions. The conditional probability of 10% is based on FEMA P-695 methodology. The absolute probability of 1% in 50 years and the conditional probability of 10% given MCE_R ground motions were used by the U.S. Geological Survey to develop the probabilistic MCE_R ground motions of ASCE/SEI 7-10. The conditional probabilities of 5% (Risk Category III structures) and 2.5% (Risk Category IV structures) represent improved reliability anticipated for structures designed with an importance factor, I_e , greater than 1.0. Although not specifically stated by ASCE/SEI 7-10 commentary, it may be presumed that Risk Category III and Risk Category IV structures have absolute collapse probability objectives that are less than 1% in 50 years (i.e., due to design using an importance factor of $I_e > 1.0$).

Engineers may need load criteria for strength design that are consistent with the requirements in this standard for situations that are not covered explicitly within this standard. They may also need to consider load criteria for special situations, as required by the client in performance-based engineering applications, in accordance with Section 1.3.1.3. In addition, groups writing standards and specifications for strength design of structural systems and elements may need to develop resistance factors that, when used with the load requirements in this standard, permit the stipulated reliability to be achieved. Such load criteria should be developed using an accepted procedure, such as that provided in Section 2.3.6, to ensure that the resulting factored design loads and load combinations are consistent with target reliabilities (or levels of performance), the common load criteria in Section 2.3.2, and existing standards and specifications governing strength design for common construction materials. Peer-reviewed statistical data for loads in this standard are provided in Table C1.3-1, adopted from Ellingwood et al. (1980, 1982) and Galambos et al. (1982). The statistics provided are the ratio of the mean, X_m , to nominal, X_n , values of the load and the coefficient of variation, COV, of a cumulative distribution function (CDF) fitted to the 90th percentile and above of the probability distribution of the load. The parameters for S (snow) are based on data for the northeastern quadrant of the United States.

The reliability of structural members can be determined through a reliability analysis, such as a Monte Carlo analysis with random variables assigned probability distributions with mean and COV values based on statistical data. Reliability analyses can also be conducted using a nonparametric hazard curve based on data. Fig. C1.3-1 provides an example of the design equation, limit state equations, and statistical variables for a compact steel flexural member designed for dead plus live load. The statistics used in this example are typical (cf. Table C1.3-1). The user should determine the appropriate probabilistic models for his or her design situation.

Fig. C1.3-1 presents an illustration of how the reliability index, β , is determined for the common case of a compact steel beam with full lateral support, in which the limit state is the formation of the first plastic hinge. The reliability depends on the ratio of nominal live to dead load in the limit state equation. For

Design Equation

$$0.9R_n = 1.2D_n + 1.6L_n$$

Limit State Equation G(X)

$$G(R, D, L) = R - D - L$$

$$G(X) = \left(\left[1.2 + 1.6 \left(\frac{L_n}{D_n} \right) \right] / 0.9 \right) X_1 - X_2 - \left(\frac{L_n}{D_n} \right) X_3$$

where

R = Strength random variable, and R_n = Nominal strength
 D = Dead load random variable, and D_n = Nominal dead load
 L = Live load random variable, and L_n = Nominal live load

$$X_1 = R/R_n, \quad X_2 = D/D_n, \quad X_3 = L/L_n;$$

Typical range of L/D_n is 0.5 to 4.0.

Statistics

Variable	Mean	Coefficient of Variation (COV)	Probability Density Function (PDF)
X_1	1.08	0.09	Lognormal
X_2	1.05	0.10	Normal
X_3	1.00	0.25	Type I

FIGURE C1.3-1 Equations and Statistics of Load and Resistance Parameters for a Monte Carlo Analysis to Determine Achieved Reliability

$L_n/D_n = 2.0$ (a typical value), the probability of failure (50-year basis) is 0.00298. The corresponding reliability index $\beta = \Phi^{-1}(1 - P_f) = 2.75$ (50-year basis), and $P_f = 6 \times 10^{-5}$ (annual basis). These reliabilities can be compared to the reliability targets in Table 1.3-1. The variation in β with L_n/D_n is very small; unlike ASD, the dead and live load factors of 1.2 and 1.6 were selected so as to properly reflect the differences in variability between dead and live load.

The system reliabilities for earthquake are different than those for other environmental hazards because the design philosophy of the standard is to prevent system collapse in the risk-targeted maximum considered earthquake (MCE_R) shaking. The R , C_d , and Ω_0 coefficients specified in Chapter 12 for seismic loading, together with the systems detailing requirements specified in the referenced standards, are intended to ensure minimum acceptable probabilities of structural collapse, given the occurrence of Maximum Considered Earthquake (MCE_R) ground shaking. As discussed in Section C11.4, for typical structures (Risk Category I and II), the conditional probability of collapse is assumed to be 10%, given the occurrence of the MCE_R . This assumption is based on significant research documented in FEMA P695 (2009). The additional collapse goals of 5% for Risk Category III and 2.5% for Risk Category IV were arrived at by assuming that the seismic fragility (probabilistic model of system strength) is described by a lognormal distribution with a logarithmic standard deviation of 0.6 and adjusting the strength of the structure by the earthquake importance factors of 1.25 and 1.5, respectively. Since collapse is a function of loading (ground shaking)

Table C1.3-1 Load Distributions and Parameters

Load	X_m/X_n	V_x	CDF
D	1.05	0.10	Normal
L	1.00	0.25	Type I
W	Site-dependent		Type I
S	0.82	0.26	Type II
E	Site-dependent		Type II

intensity, still lower, but nonnegligible probabilities of collapse also exist at design shaking levels. The collapse risk for design earthquake shaking is approximately 2.5%, 1%, and 0.5% for Risk Category II, III, and IV structures.

This standard also seeks to protect against local failure that does not result in global collapse but could result in injury risk to a few persons. Chapter 16 of the standard defines structural elements according to their criticality as critical, ordinary, and noncritical, where critical elements can lead to global collapse, ordinary elements to endangerment of a limited number of lives, and noncritical elements do not have safety consequences. For ordinary elements in Risk Category II structures, the standard accepts a 25% probability of failure given MCE_R shaking (approximately 10% probability of failure for design earthquake shaking). Failure probabilities for ordinary elements in Risk Category III and IV structures are, respectively, 15% and 9% for MCE_R shaking and 4% and 2% for DE shaking. It is anticipated that the failure probabilities for anchorage of rigid nonstructural components attached at grade to the structure may be in the same range as the probabilities for ordinary elements. However, the uncertainties associated with the reliability for anchorage of rigid nonstructural components that are elevated within a structure are much higher than for structural elements because the methods used to characterize the strength demands on nonstructural components are more approximate than those used for overall building demands, and appropriate reliability levels have not yet been established for them. Furthermore, demands on anchorage of flexible nonstructural components and distributions are significantly more complex, especially when points of attachment of the nonstructural components are elevated within a structure and need to consider both inertial effects and relative displacements. Future study should seek to evaluate nonstructural reliabilities in a rigorous manner, and if consensus can be achieved regarding the appropriate reliability levels for anchorage, to adjust the design procedure for anchorage of these components to achieve these appropriate reliabilities.

It is important to note that provision of adequate strength is not sufficient to ensure proper performance. Considerations of serviceability and structural integrity are also important. Use of the alternative procedures of Section 1.3.1.3 is not intended as an alternative to the requirements of Sections 1.3.2, 1.3.3, 1.3.4, 1.3.5, 1.3.6, or 1.4 of this standard.

The requirements of this standard and its companion referenced standards are intended to provide protection against structural failure. They are also intended to provide property and economic protection for small events, to the extent practical, as well as improve the probability that critical facilities will be functional after severe storms, earthquakes, and similar events. Although these goals are an important part of the requirements of this standard, at the present time there is no documentation of the reliability intended with respect to these goals. Consequently, Tables 1.3-1 and 1.3-2 address safety considerations only.

Compliance with Section 1.3.1.3 may be demonstrated by analysis, testing, or a combination of both methods. It is important to recognize that there is uncertainty as to whether the performance objectives tabulated in Tables 1.3-1 and 1.3-2 can be achieved. There is inherent uncertainty associated with prediction of the intensity of loading that a structure will experience, the actual strength of materials incorporated in construction, the quality of construction, and the condition of the structure at the time of loading. Whether testing, analysis, or a combination of these is used, provision should be made to account for these uncertainties and to ensure that the probability of poor performance is acceptably low. See Ellingwood et al. (1982) and Galambos et al. (1982) for estimates of such uncertainties.

Rigorous methods of reliability analysis can be used to demonstrate that the reliability of a design achieves the targets indicated in Tables 1.3-1 and 1.3-2; a simple illustration of such a method is provided in C1.3-1. While such analyses would certainly constitute an acceptable approach to satisfy the requirements of Section 1.3.1.3, these may not be the only acceptable approaches. Consensus bodies or other standards developing organizations may develop guideline documents that provide alternative performance-based design methods or alternate prescriptive procedures that meet or exceed the reliabilities stated in this section.

Since most building officials and other Authorities Having Jurisdiction do not have the expertise necessary to judge the adequacy of designs justified using the Section 1.3.1.3 procedures, independent peer review is an essential part of this process. Peer review can help to reduce the potential that the design professional of record will overlook or misinterpret one or more potential behaviors that could result in poor performance. Peer review can also help to establish that an appropriate standard of care was adhered to during the design. For peer review to be effective, the reviewers must have the appropriate expertise and understanding of the types of structures, loading, analysis methods, and testing used in the procedures.

The target reliabilities listed in Tables 1.3-1 for members and connections and those in Tables 1.3-2 and 1.3-3 for structural systems are included in this standard specifically for application to performance-based procedures for individual projects that are peer-reviewed by experts in the field. For several reasons, these target reliabilities are not intended to be compared to reliability indices developed by material specification groups for general structural applications. For example, reliability indices for some materials are based on testing of small coupons of the material supplemented by factors to account for scaling up to structural-sized elements, whereas other materials test full-sized structural members as the foundation for this analysis. Additionally, some reliability analyses use default lognormal data distribution assumptions, and others use distributional forms applicable to the material and to each load combination of interest.

C1.3.1.3.2 Testing. Laboratory testing of materials and components constructed from those materials is an essential part of the process of validating the performance of structures and nonstructural components under load. Design resistances specified in the industry standards used with the strength procedures of Section 1.3.1.1 and the allowable stress procedures of Section 1.3.1.2 are based on extensive laboratory testing and many years of experience with the performance of structures designed using these standards in real structures. Similarly, analytical modeling techniques commonly used by engineers to predict the behavior of these systems have been benchmarked and validated against laboratory testing. Similar benchmarking of resistance, component performance, and analytical models is essential when performance-based procedures are used. Where systems and components that are within the scope of the industry standards are used in a design, analytical modeling of these systems and components and their resistances should be conducted in accordance with these standards and industry practice, unless new data and testing suggest that other assumptions are more appropriate. Where new systems, components, or materials are to be used, laboratory testing must be performed to indicate appropriate modeling assumptions and resistances.

No single protocol is appropriate for use in laboratory testing of structural and nonstructural components. The appropriate number and types of tests that should be performed depend on the type of loading the component will be subjected to, the

complexity of the component's behavior, the failure modes it may exhibit, the consequences of this failure, and the variability associated with the behavior. Resistances should be selected to provide an acceptably low probability of unacceptable performance. Commentary to Chapter 2 provides guidance on the calculation of load and resistance factors that may be used for this purpose, when LRFD procedures are used.

Regardless of the means used to demonstrate acceptable performance, testing should be sufficient to provide an understanding of the probable mean value and variability of resistance or component performance. For materials or components that exhibit significant variability in behavior, as a result either of workmanship, material variation, or brittle modes of behavior, a very large number of tests may be required to properly characterize both the mean values and dispersion. It is seldom possible to conduct such a large number of tests as part of an individual project. Therefore, for reasons of practicality, this standard permits a small number of tests, with the number based on the observed variability. Users are cautioned to conduct tests on material that is representative of that expected to be used in the specific project and that all significant sources of variability are included in the test samples. When high variability is observed in these test data, the minimum requirement of six tests is not adequate to establish either the true mean or the variability with confidence, and appropriate caution should be used when developing component resistance or performance measures based on this limited testing. This is a primary reason why the procedures of this section are limited to use on the specific projects being analyzed (i.e., they are not "portable" to similar projects) and that data from these tests are not intended as a means of obtaining prequalification of new systems, materials, or components for broad application.

Some industries and industry standards have adopted standard protocols and procedures for qualification testing. For example, AISC 341, Chapter K, specifies the required testing for qualification of connections used in certain steel seismic force-resisting systems. The wood structural panel industry has generally embraced the testing protocols developed by the Consortium of Universities for Research in Earthquake Engineering project (Krawinkler et al. 2002). When a material, component, or system is similar to those for which such an industry standard exists, the industry standard should be used, unless it can be demonstrated to the satisfaction of peer review and the Authority Having Jurisdiction that more appropriate results will be attained by using alternative procedures and protocols.

When data from Section 1.3.1.3.2 testing is used to characterize a variable within the reliability analysis, sample sizes shall be sufficient to define the mean and coefficient of variation of the test results within specific confidence bounds determined by the significance of that variable in the reliability analysis. While most testing being conducted in accordance with Section 1.3.1.3.2 will be used primarily to confirm or supplement engineering analyses, it is possible that test data will also be used to characterize one of the random variables that are part of the reliability analysis. Because each variable in the reliability analysis will influence the final computed reliability index, each test-based variable must be subjected to a reasonable amount of statistical rigor. For very small sample sizes, it is not possible to define the mean or the standard deviation precisely. However, there are well-established methods to compute confidence bounds on those parameters. For example, a conservative estimate of the mean value might be the lower 75% confidence bound.

C1.3.2 Serviceability. In addition to strength limit states, buildings and other structures must also satisfy serviceability

limit states that define functional performance and behavior under loads normally experienced during the lifetime of the structure or during a time defined specifically for a project or a particular limit state. Serviceability limit states include such items as deflection and vibration. In the United States, strength limit states have traditionally been specified in building codes because they control the safety of the structure. Serviceability limit states, however, are usually noncatastrophic, define a level of quality of the structure or element, and are a matter of judgment as to their application. Serviceability limit states involve the perceptions and expectations of the owner or user and are a contractual matter between the owner or user and the designer and builder. It is for these reasons, and because the benefits are often subjective and difficult to define or quantify, that serviceability limit states for the most part are not included within the model U.S. building codes, with several notable exceptions, such as member deflection limits. In some cases, material design standards provide serviceability limit states for structural elements composed of their material.

The fact that serviceability limit states are usually not codified should not diminish their importance. Exceeding a serviceability limit state in a building or other structure usually means that its function is disrupted or impaired because of local minor damage or deterioration or because of occupant discomfort or annoyance. Therefore, this section states that serviceability limit states and the service loads associated with those limit states should be defined in the project design criteria, which would often be developed in consultation with the owner of the building or other structure. Appendix C and its commentary provide guidance to the designer on developing serviceability design criteria.

Service loads can vary significantly from the design loads specified in this standard. Often the service loads are dependent on the specific serviceability limit state being investigated. For example, beam deflection for a stiffness serviceability limit state has typically been evaluated using the live load is specified in this standard without a load factor applied to it. While the live load used for evaluating floor vibration caused by footfall has commonly been taken as an estimated average of the actual live load present, often significantly less than the design live load is specified in this standard.

C1.3.3 Functionality. Structures in Risk Category IV are intended to have some measure of protection against damage to the structure and designated nonstructural systems that would preclude the facility from resuming its intended function following the design environmental hazard. For example, in Chapter 13 of the standard, nonstructural systems assigned to Risk Category IV must be tested or verified by analysis to be rugged enough to retain their pre-earthquake function following the design earthquake. There are additional requirements in the seismic chapters, 11 through 23, to limit structural damage and drift to preserve function of the structure. Since the provisions of this standard require that structures and nonstructural components be designed to remain essentially elastic under most other environmental hazards, preservation of function is generally provided.

When a performance-based design is elected over the prescriptive design procedures in the standard, the registered design professional should confirm that the structure has sufficient strength and stiffness to not incur damage during the design environmental hazard event that would prevent the facility from resuming its intended function, or in some cases or hazards, functioning during the design hazard.

Because the nature of function preservation is very broad and encompasses many different structural components and systems subjected to various hazards, there are not specific reliability

targets. For that reason, the terms “reasonable probability” and “adequate structural strength” as applied to structural systems are used to indicate that the application of Section 1.1.3 is not an absolute target value. What constitutes reasonable probability depends on many factors, including the recognition that the fragilities of structural systems to ensure function are not well established and should be agreed upon by the user, the client, and if applicable, the Authority Having Jurisdiction and the peer reviewers.

The designated nonstructural systems should be composed of components that have adequate strength, stiffness, and ruggedness and are adequately attached to the structure so that they do not incur damage sufficient to prohibit the function of the system within a specified period of time as the facility is brought back online. This functionality can be demonstrated through analysis or through physical testing of the nonstructural system or components.

Designated nonstructural systems may vary between structures based on the function of the facility. However, systems that are essential to Life Safety are commonly accepted as needed function preservation. Such systems may include fire detection and suppression systems, emergency exit lighting, and systems that contain explosive, toxic, or highly toxic materials. The specific systems and components that should be considered part of the “designated nonstructural system” should be determined by the user and may require approval by the Authority Having Jurisdiction and, if part of the design, the peer reviewers.

The requirements of this section are not to preclude damage to the structural elements or nonstructural components. In fact, there might be considerable damage to nonessential nonstructural systems and some indications of inelastic deformation of the structure. There should be no structural damage that would indicate the structure is unsafe to support the loads with reliability similar to that required to support it during normal operation. There should be no damage to the nonstructural components that prevents function, such as blocked egress routes. Designated nonstructural systems may have cosmetic damage, but the components can function as they did before the hazard.

The statement that the provisions within this standard are deemed to comply with this section means that through following the prescriptive provisions contained herein, the design professional should be able to provide the intended function preservation performance of Risk Category IV structures in a reasonable period of time consistent with the current state of the practice. In some chapters of this standard, there are other terms that refer to essentially the same components as “Designated Nonstructural System,” “Designated Seismic System” in Chapter 13 and “Critical Equipment and Systems” in Chapter 6 are two examples. In those cases, the hazard chapter-specific terms are retained to allow for the hazard-specific requirements to be implemented.

C1.3.4 Self-Straining Forces and Effects. Indeterminate structures that experience dimensional changes develop self-straining forces and effects. Examples include moments in rigid frames that undergo differential foundation settlements, pretensioning or post-tensioning forces as well as any relaxation or loss of such forces sufficient to affect structural performance, and shear forces in bearing walls that support concrete slabs that shrink. Unless provisions are made for self-straining forces and effects, stresses in structural elements, either alone or in combination with stresses from external loads, can be high enough to cause structural damage.

In many cases, the magnitude of self-straining forces can be anticipated by analyses of expected shrinkage, temperature fluctuations, foundation movement, and so forth. However, it is not

always practical to calculate the magnitude of self-straining forces. Designers often specify relief joints, suitable framing systems, or other details to minimize self-straining forces and effects.

C1.3.7 Fire Resistance. Where there is no applicable building code (the building code under which the structure is designed), the structural fire resistance provisions of the International Building Code should be applied. Appendix E provides performance-based design procedures for alternative means specified in Section 1.3.1.

C1.4 GENERAL STRUCTURAL INTEGRITY

Sections 1.4.1 through 1.4.4 present minimum strength criteria intended to ensure that all structures are provided with minimum interconnectivity of their elements and that a complete lateral force-resisting system is present with sufficient strength to provide for stability under gravity loads and nominal lateral forces that are independent of design wind, seismic, or other anticipated loads. Conformance with these criteria provides structural integrity for normal service and minor unanticipated events that may reasonably be expected to occur throughout their lifetimes. For many structures that house large numbers of persons, or that house functions necessary to protect the public safety or occupancies that may be the subject of intentional sabotage or attack, more rigorous protection should be incorporated into designs than provided by these sections. For such structures, additional precautions can and should be taken in the design of structures to limit the effects of local collapse and to prevent or minimize progressive collapse in accordance with the procedures of Section 2.5, as charged by Section 1.4. Progressive collapse is defined as the spread of an initial local failure from element to element, resulting eventually in the collapse of an entire structure or a disproportionately large part of it.

Some authors have defined resistance to progressive collapse to be the ability of a structure to accommodate, with only local failure, the notional removal of any single structural member. Aside from the possibility of further damage that uncontrolled debris from the failed member may cause, it appears prudent to consider whether the abnormal event will fail only a single member.

Because accidents, misuse, and sabotage are normally unforeseeable events, they cannot be defined precisely. Likewise, general structural integrity is a quality that cannot be stated in simple terms. It is the purpose of Section 1.4 and the commentary to direct attention to the problem of local collapse, present guidelines for handling it that will aid the design engineer, and promote consistency of treatment in all types of structures and in all construction materials. ASCE does not intend, at this time, for this standard to establish specific events to be considered during design or for this standard to provide specific design criteria to minimize the risk of progressive collapse.

Accidents, Misuse, Sabotage, and Their Consequences. In addition to unintentional or willful misuse, some of the incidents that may cause local collapse (Leyendecker et al. 1976) are explosions caused by ignition of gas or industrial liquids, boiler failures, vehicle impact, impact of falling objects, effects of adjacent excavations, gross construction errors, very high winds such as tornadoes, and sabotage. Generally, such abnormal events would not be a part of normal design considerations. The distinction between general collapse and limited local collapse can best be made by example as follows.

General Collapse. The immediate, deliberate demolition of an entire structure by phased explosives is an obvious instance of general collapse. Also, the failure of one column in a one-, two-, three-, or possibly even four-column structure could precipitate

general collapse because the local failed column is a significant part of the total structural system at that level. Similarly, the failure of a major bearing element in the bottom story of a two- or three-story structure might cause general collapse of the whole structure. Such collapses are beyond the scope of the provisions discussed herein. There have been numerous instances of general collapse that have occurred as the result of such events as bombing, landslides, and floods.

Limited Local Collapse. An example of limited local collapse would be the containment of damage to adjacent bays and stories following the destruction of one or two neighboring columns in a multibay structure. The restriction of damage to portions of two or three stories of a higher structure following the failure of a section of bearing wall in one story is another example.

Examples of General Collapse

Ronan Point. A prominent case of local collapse that progressed to a disproportionate part of the whole building (and is thus an example of the type of failure of concern here) was the Ronan Point disaster, which brought the attention of the profession to the matter of general structural integrity in buildings. Ronan Point was a 22-story apartment building of large, precast-concrete, load-bearing panels in Canning Town, England. In March 1968, a gas explosion in an 18th-story apartment blew out a living room wall. The loss of the wall led to the collapse of the whole corner of the building. The apartments above the 18th story, suddenly losing support from below and being insufficiently tied and reinforced, collapsed one after the other. The falling debris ruptured successive floors and walls below the 18th story, and the failure progressed to the ground. Better continuity and ductility might have reduced the amount of damage at Ronan Point.

Another example is the failure of a one-story parking garage reported in Granstrom and Carlsson (1974). Collapse of one transverse frame under a concentration of snow led to the later progressive collapse of the whole roof, which was supported by 20 transverse frames of the same type. Similar progressive collapses are mentioned in Seltz-Petrash (1979).

Alfred P. Murrah Federal Building. On April 19, 1995, a truck containing approximately 4,000 lb of fertilizer-based explosive (ammonium nitrate/fuel oil) was parked near the sidewalk next to the nine-story reinforced concrete office building (Weidinger 1994; *Engrg. News Rec.* 1995; Longinow 1995; Glover 1996). The side facing the blast had corner columns and four other perimeter columns. The blast shock wave disintegrated one of the 20- × 36-in. (508 × 915 mm) perimeter columns and caused brittle failures of two others. The transfer girder at the third level above these columns failed, and the upper-story floors collapsed in a progressive fashion. Approximately 70% of the building experienced dramatic collapse. One hundred sixty-eight people died, many of them as a direct result of progressive collapse. Damage might have been less had this structure not relied on transfer girders for support of upper floors, if there had been better detailing for ductility and greater redundancy, and if there had been better resistance for uplift loads on floor slabs.

There are a number of factors that contribute to the risk of damage propagation in modern structures (Breen 1976). Among them are the following:

1. There is an apparent lack of general awareness among engineers that structural integrity against collapse is important enough to be regularly considered in design.
2. To have more flexibility in floor plans and to keep costs down, interior walls and partitions are often non-load-bearing and hence may be unable to assist in containing damage.

3. In attempting to achieve economy in structure through greater speed of erection and less site labor, systems may be built with minimum continuity, ties between elements, and joint rigidity.
4. Unreinforced or lightly reinforced load-bearing walls in multistory structures may also have inadequate continuity, ties, and joint rigidity.
5. In roof trusses and arches, there may not be sufficient strength to carry the extra loads or sufficient diaphragm action to maintain lateral stability of the adjacent members if one collapses.
6. In eliminating excessively large safety factors, code changes over the past several decades have reduced the large margin of safety inherent in many older structures. The use of higher strength materials permitting more slender sections compounds the problem in that modern structures may be more flexible and sensitive to load variations and, in addition, may be more sensitive to construction errors.

Experience has demonstrated that the principle of taking precautions in design to limit the effects of local collapse is realistic and can be satisfied economically. From a public safety viewpoint, it is reasonable to expect all multistory structures to possess general structural integrity comparable to that of properly designed, conventionally framed structures (Breen 1976; Burnett 1975).

Design Alternatives. There are a number of ways to obtain resistance to progressive collapse. In Ellingwood and Leyendecker (1978), a distinction is made between direct and indirect design, and the following approaches are defined:

Direct Design: Explicit consideration of resistance to progressive collapse during the design process through either

Alternate Path Method: A method that allows local failure to occur but seeks to provide alternate load paths so that the damage is absorbed and major collapse is averted.

Specific Local Resistance Method: A method that seeks to provide sufficient strength to resist failure from accidents or misuse.

Indirect Design: Implicit consideration of resistance to progressive collapse during the design process through the provision of minimum levels of strength, continuity, and ductility.

The general structural integrity of a structure may be tested by analysis to ascertain whether alternate paths around hypothetically collapsed regions exist. Alternatively, alternate path studies may be used as guides for developing rules for the minimum levels of continuity and ductility needed to apply the indirect design approach to enhance general structural integrity. Specific local resistance may be provided in regions of high risk because it may be necessary for some element to have sufficient strength to resist abnormal loads for the structure as a whole to develop alternate paths. Specific suggestions for the implementation of each of the defined methods are contained in Ellingwood and Leyendecker (1978).

Guidelines for the Provision of General Structural Integrity. Generally, connections between structural components should be ductile and have a capacity for relatively large deformations and energy absorption under the effect of abnormal conditions. This criterion is met in many different ways, depending on the structural system used. Details that are appropriate for resistance to moderate wind loads and seismic loads often provide sufficient ductility. In 1999, ASCE issued a state of practice report that is a good introduction to the complex field of blast-resistant design (ASCE 1999).

Work with large precast panel structures (Schultz et al. 1977; PCI Committee on Precast Bearing Walls 1976; Fintel and Schultz 1979) provides an example of how to cope with the problem of general structural integrity in a building system that is inherently discontinuous. The provision of ties combined with careful detailing of connections can overcome difficulties associated with such a system. The same kind of methodology and design philosophy can be applied to other systems (Fintel and Annamalai 1979). The *ACI Building Code Requirements for Structural Concrete* (ACI 2014) includes such requirements in Section 4.10.

There are a number of ways of designing for the required integrity to carry loads around severely damaged walls, trusses, beams, columns, and floors. A few examples of design concepts and details are

1. Good plan layout. An important factor in achieving integrity is the proper plan layout of walls and columns. In bearing-wall structures, there should be an arrangement of interior longitudinal walls to support and reduce the span of long sections of crosswall, thus enhancing the stability of individual walls and of the structures as a whole. In the case of local failure, this reduction will also decrease the length of wall likely to be affected.
2. Provide an integrated system of ties among the principal elements of the structural system. These ties may be designed specifically as components of secondary load-carrying systems, which often must sustain very large deformations during catastrophic events.
3. Returns on walls. Returns on interior and exterior walls make them more stable.
4. Changing directions of span of floor slab. Where a one-way floor slab is reinforced to span, with a low safety factor, in its secondary direction if a load-bearing wall is removed, the collapse of the slab will be prevented and the debris loading of other parts of the structure will be minimized. Often, shrinkage and temperature steel will be enough to enable the slab to span in a new direction.
5. Load-bearing interior partitions. The interior walls must be capable of carrying enough load to achieve the change of span direction in the floor slabs.
6. Catenary action of floor slab. Where the slab cannot change span direction, the span will increase if an intermediate supporting wall is removed. In this case, if there is enough reinforcement throughout the slab and enough continuity and restraint, the slab may be capable of carrying the loads by catenary action, though very large deflections will result.
7. Beam action of walls. Walls may be assumed to be capable of spanning an opening if sufficient tying steel at the top and bottom of the walls allows them to act as the web of a beam with the slabs above and below acting as flanges (Schultz et al. 1977).
8. Redundant structural systems. A secondary load path (e.g., an upper-level truss or transfer girder system that allows the lower floors of a multistory building to hang from the upper floors in an emergency) allows framing to survive removal of key support elements.
9. Ductile detailing. Avoid low-ductility detailing in elements that might be subject to dynamic loads or very large distortions during localized failures (e.g., consider the implications of shear failures in beams or supported slabs under the influence of building weights falling from above).
10. Provide additional reinforcement to resist blast and load reversal when blast loads are considered in design (ASCE 1997).
11. Consider the use of compartmentalized construction in combination with special moment-resisting frames (as defined in FEMA 1997) in the design of new buildings when considering blast protection.

Although not directly adding structural integrity for the prevention of progressive collapse, the use of special, nonfrangible glass for fenestration can greatly reduce risk to occupants during exterior blasts (ASCE 1997). To the extent that nonfrangible glass isolates a building's interior from blast shock waves, it can also reduce damage to interior framing elements (e.g., supported floor slabs could be made to be less likely to fail due to uplift forces) for exterior blasts.

C1.5 CLASSIFICATION OF BUILDINGS AND OTHER STRUCTURES

C1.5.1 Risk Categorization. In the 2010 edition of this standard, a new Table 1.5-2 was added that consolidates the various importance factors specified for the several type of loads throughout the standard in one location. This change was made to facilitate the process of finding values of these factors. Simultaneously with this addition, the importance factors for wind loads were deleted as changes to the new wind hazard maps adopted by the standard incorporated consideration of less probable design winds for structures assigned to higher Risk Categories, negating the need for separate importance factors. Further commentary on this issue may be found in the commentary to Chapter 26.

The Risk Categories in Table 1.5-1 are used to relate the criteria for maximum environmental loads or distortions specified in this standard to the consequence of the loads being exceeded for the structure and its occupants. For many years, this standard used the term "Occupancy Category," as have the building codes. However, the term "occupancy" as used by the building codes relates primarily to issues associated with fire and Life Safety protection, as opposed to the risks associated with structural failure. The term Risk Category was adopted in place of the older Occupancy Category in the 2010 edition of the standard to distinguish between these two considerations. The Risk Category numbering is unchanged from that in the previous editions of the standard (ASCE 7-98, -02, and -05), but the criteria for selecting a category have been generalized with regard to structure and occupancy descriptions. The reason for this generalization is that the acceptable risk for a building or structure is an issue of public policy, rather than purely a technical one. Model building codes such as the International Building Code (ICC 2009) and NFPA-5000 (NFPA 2006) contain prescriptive lists of building types by Occupancy Category. Individual communities can alter these lists when they adopt local codes based on the model code, and individual owners or operators can elect to design individual buildings to higher Occupancy Categories based on personal risk management decisions. Classification continues to reflect a progression of the anticipated seriousness of the consequence of failure from lowest risk to human life (Risk Category I) to the highest (Risk Category IV). Elimination of the specific examples of buildings that fall into each category has the benefit that it eliminates the potential for conflict between the standard and locally adopted codes and also provides individual communities and development teams the flexibility to interpret acceptable risk for individual projects.

Historically, the building codes and the standard have used a variety of factors to determine the Occupancy Category of a building. These factors include the total number of persons who

would be at risk were failure to occur, the total number of persons present in a single room or occupied area, the mobility of the occupants and their ability to cope with dangerous situations, the potential for release of toxic materials, and the loss of services vital to the welfare of the community.

Risk Category I structures generally encompass buildings and structures that normally are unoccupied and that would result in negligible risk to the public should they fail. Structures typically classified in this category have included barns, storage shelters, gatehouses, and similar small structures. Risk Category II includes the vast majority of structures, including most residential, commercial, and industrial buildings and has historically been designated as containing all those buildings and structures not specifically classified as conforming to another category.

Risk Category III includes buildings and structures that house a large number of persons in one place, such as theaters, lecture halls, and similar assembly uses and buildings with persons having limited mobility or ability to escape to a safe haven in the event of failure, including elementary schools, prisons, and small health-care facilities. This category has also included structures associated with utilities required to protect the health and safety of a community, including power-generating stations and water-treatment and sewage-treatment plants. It has also included structures housing hazardous substances, such as explosives or toxins, which if released in quantity could endanger the surrounding community, such as structures in petrochemical process facilities containing large quantities of H_2S or ammonia.

Failures of power plants that supply electricity on the national grid can cause substantial economic losses and disruption to civilian life when their failures can trigger other plants to go offline in succession. The result can be massive and potentially extended power outage, shortage, or both that lead to huge economic losses because of idled industries and a serious disruption of civilian life because of inoperable subways, road traffic signals, and so forth. One such event occurred in parts of Canada and the northeastern United States in August 2003.

Failures of water- and sewage-treatment facilities can cause disruption to civilian life because these failures can cause large-scale (but mostly non-life-threatening) public health risks caused by the inability to treat sewage and to provide drinking water.

Failures of major telecommunication centers can cause disruption to civilian life by depriving users of access to important emergency information (using radio, television, and phone communication) and by causing substantial economic losses associated with widespread interruption of business.

Risk Category IV has traditionally included structures the failure of which would inhibit the availability of essential community services necessary to cope with an emergency situation. Buildings and structures typically grouped in Risk Category IV include hospitals, police stations, fire stations, emergency communication centers, and similar uses.

Ancillary structures required for the operation of Risk Category IV facilities during an emergency also are included in this Risk Category. When deciding whether an ancillary structure or a structure that supports such functions as fire suppression is in Risk Category IV, the design professional must decide whether failure of the subject structure will adversely affect the essential function of the facility. In addition to essential facilities, buildings and other structures containing extremely hazardous materials have been added to Risk Category IV to recognize the potentially devastating effect a release of extremely hazardous materials may have on a population.

The criteria that have historically been used to assign individual buildings and structures to Occupancy Categories have not

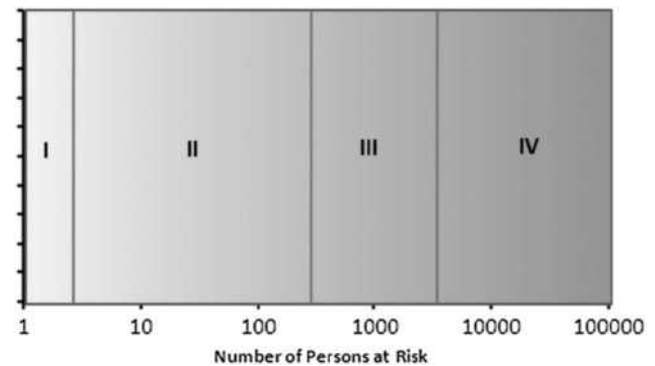


FIGURE C1.5-1 Approximate Relationship between Number of Lives Placed at Risk by a Failure and Occupancy Category

been consistent and sometimes have been based on considerations that are more appropriate to fire and Life Safety than to structural failure. For example, university buildings housing more than a few hundred students have been placed into a higher Risk Category than office buildings housing the same number of persons.

A rational basis should be used to determine the Risk Category for structural design, which is primarily based on the number of persons whose lives would be endangered or whose welfare would be affected in the event of failure. Fig. C1.5-1 illustrates this concept.

“Lives at risk” pertains to the number of people at serious risk of life loss given a structural failure. The Risk Category classification is not the same as the building code occupancy capacity, which is mostly based on risk to life from fire. The lives at risk from a structural failure include persons who may be outside the structure in question who are nonetheless put at serious risk by failure of the structure. From this concept, emergency recovery facilities that serve large populations, even though the structure might shelter relatively few people, are moved into the higher Risk Categories.

When determining the population at risk, consideration should also be given to longer term risks to life than those created during a structural failure. The failure of some buildings and structures, or their inability to function after a severe storm, earthquake, or other disaster, can have far-reaching impact. For example, loss of functionality in one or more fire stations could inhibit the ability of a fire department to extinguish fires, allowing fires to spread and placing many more people at risk. Similarly, the loss of function of a hospital could prevent the treatment of many patients over a period of months.

In Chapters 7, 10, and 11, importance factors are presented for the four Risk Categories identified. The specific importance factors differ according to the statistical characteristics of the environmental loads and the manner in which the structure responds to the loads. The principle of requiring more stringent loading criteria for situations in which the consequence of failure may be severe has been recognized in previous versions of this standard by the specification of mean recurrence interval maps for wind speed and ground snow load.

This section now recognizes that there may be situations when it is acceptable to assign multiple Risk Categories to a structure based on use and the type of load condition being evaluated. For instance, there are circumstances when a structure should appropriately be designed for wind loads with importance factors greater than one, but would be penalized unnecessarily if designed for seismic loads with importance factors greater than

one. An example would be a hurricane shelter in a low seismic area. The structure would be classified in Risk Category IV for wind design and in Risk Category II for seismic design.

C1.5.3 Toxic, Highly Toxic, and Explosive Substances. A common method of categorizing structures storing toxic, highly toxic, or explosive substances is by the use of a table of exempt amounts of these materials (EPA 1999b; International Code Council 2000). These references and others are sources of guidance on the identification of materials of these general classifications. A drawback to the use of tables of exempt amounts is the fact that the method cannot handle the interaction of multiple materials. Two materials may be exempt because neither poses a risk to the public by themselves but may form a deadly combination if combined in a release. Therefore, an alternate and superior method of evaluating the risk to the public of a release of a material is by a hazard assessment as part of an overall risk management plan (RMP).

Buildings and other structures containing toxic, highly toxic, or explosive substances may be classified as Risk Category II structures if it can be demonstrated that the risk to the public from a release of these materials is minimal. Companies that operate industrial facilities typically perform hazard and operability (HAZOP) studies, conduct quantitative risk assessments, and develop risk management and emergency response plans. Federal regulations and local laws mandate many of these studies and plans (EPA 1999a). Additionally, many industrial facilities are located in areas remote from the public and have restricted access, which further reduces the risk to the public.

The intent of Section 1.5.2 is for the RMP and the facility's design features that are critical to the effective implementation of the RMP to be maintained for the life of the facility. The RMP and its associated critical design features must be reviewed on a regular basis to ensure that the actual condition of the facility is consistent with the plan. The RMP also should be reviewed whenever consideration is given to the alteration of facility features that are critical to the effective implementation of the RMP.

The RMP generally deals with mitigating the risk to the general public. Risk to individuals outside the facility storing toxic, highly toxic, or explosive substances is emphasized because plant personnel are not placed at as high a risk as the general public because of the plant personnel's training in the handling of toxic, highly toxic, or explosive substances and because of the safety procedures implemented inside the facilities. When these elements (trained personnel and safety procedures) are not present in a facility, then the RMP must mitigate the risk to the plant personnel in the same manner as it mitigates the risk to the general public.

As a result of the prevention program portion of an RMP, buildings and other structures normally falling into Risk Category III may be classified into Risk Category II if means (e.g., secondary containment) are provided to contain the toxic, highly toxic, or explosive substances in the case of a release. To qualify, secondary containment systems must be designed, installed, and operated to prevent migration of harmful quantities of toxic, highly toxic, or explosive substances out of the system to the air, soil, groundwater, or surface water at any time during the use of the structure. This requirement is not to be construed as requiring a secondary containment system to prevent a release of any toxic, highly toxic, or explosive substance into the air. By recognizing that secondary containment shall not allow releases of "harmful" quantities of contaminants, this standard acknowledges that there are substances that might contaminate groundwater but do not produce a sufficient concentration of toxic, highly toxic, or explosive substances during a vapor release to

constitute a health or safety risk to the public. Because it represents the "last line of defense," secondary containment does not qualify for the reduced classification.

If the beneficial effect of secondary containment can be negated by external forces, such as the overtopping of dike walls by floodwaters or the loss of liquid containment of an earthen dike because of excessive ground displacement during a seismic event, then the buildings or other structures in question may not be classified into Risk Category II. If the secondary containment is to contain a flammable substance, then implementation of a program of emergency response and preparedness combined with an appropriate fire suppression system would be a prudent action associated with a Risk Category II classification. In many jurisdictions, such actions are required by local fire codes.

Also as the result of the prevention program portion of an RMP, buildings and other structures containing toxic, highly toxic, or explosive substances also could be classified as Risk Category II for hurricane wind loads when mandatory procedures are used to reduce the risk of release of toxic, highly toxic, or explosive substances during and immediately after these predictable extreme loadings. Examples of such procedures include draining hazardous fluids from a tank when a hurricane is predicted or, conversely, filling a tank with fluid to increase its buckling and overturning resistance. As appropriate to minimize the risk of damage to structures containing toxic, highly toxic, or explosive substances, mandatory procedures necessary for the Risk Category II classification should include preventative measures, such as the removal of objects that might become airborne missiles in the vicinity of the structure.

In previous editions of ASCE 7, the definitions of "hazardous" and "extremely hazardous" materials were not provided. Therefore, the determination of the distinction between hazardous and extremely hazardous materials was left to the discretion of the Authority Having Jurisdiction. The change to the use of the terms "toxic" and "highly toxic" based on definitions from Federal law (29 CFR 1910.1200 Appendix A with Amendments as of February 1, 2000) has corrected this problem.

Because of the highly quantitative nature of the definitions for toxic and highly toxic found in 29 CFR 1910.1200 Appendix A, the General Provisions Task Committee felt that the definitions found in federal law should be directly referenced instead of repeated in the body of ASCE 7. The definitions found in 29 CFR 1910.1200 Appendix A are repeated in the following text for reference.

Highly Toxic. A chemical falling within any of the following categories:

1. A chemical that has a median lethal dose [LD(50)] of 50 mg or less per kilogram of body weight when administered orally to albino rats weighing between 200 and 300 g each.
2. A chemical that has a median lethal dose [LD(50)] of 200 mg or less per kilogram of body weight when administered by continuous contact for 24 hr (or less if death occurs within 24 hr) with the bare skin of albino rabbits weighing between 2 and 3 kg each.
3. A chemical that has a median lethal concentration [LC(50)] in air of 200 parts per million by volume or less of gas or vapor, or 2 mg per liter or less of mist, fume, or dust, when administered by continuous inhalation for 1 hr (or less if death occurs within 1 hr) to albino rats weighing between 200 and 300 g each.

Toxic. A chemical falling within any of the following categories:

1. A chemical that has a median lethal dose [LD(50)] of more than 50 mg per kg, but not more than 500 mg per kg of

body weight when administered orally to albino rats weighing between 200 and 300 g each.

2. A chemical that has a median lethal dose [LD(50)] of more than 200 mg per kilogram, but not more than 1,000 mg per kilogram of body weight when administered by continuous contact for 24 hr (or less if death occurs within 24 hr) with the bare skin of albino rabbits weighing between 2 and 3 kg each.
3. A chemical that has a median lethal concentration [LC (50)] in air of more than 200 parts per million but not more than 2,000 parts per million by volume of gas or vapor, or more than 2 mg per liter but not more than 20 mg per liter of mist, fume, or dust, when administered by continuous inhalation for 1 hr (or less if death occurs within 1 hr) to albino rats weighing between 200 and 300 g each.

C1.7 LOAD TESTS

No specific method of test for completed construction has been given in this standard because it may be found advisable to vary the procedure according to conditions. Some codes require the construction to sustain a superimposed load equal to a stated multiple of the design load without evidence of serious damage. Others specify that the superimposed load shall be equal to a stated multiple of the live load plus a portion of the dead load. Limits are set on maximum deflection under load and after removal of the load. Recovery of at least three-quarters of the maximum deflection, within 24 hr after the load is removed, is a common requirement (ACI 2014).

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